ANONYMOUS TECHNICAL REPORT

HW4: Spectrum of digital signals

One-sided PSD estimation, bit-pattern identification and reproducible verification.

Three supplied 200,000-bit files are converted to trapezoidal NRZ and analyzed with the averaged periodogram. Six PSD plots cover 0-50 GHz at K = 10 and K = 100. The analysis checks the estimator's power normalization and audits the actual bit files before interpreting their spectra.

File	Evidence-based classification
Pattern 1	Consistent with independent random NRZ.
Pattern 2	The intended 8b/10b-like example; low-frequency shaping is clear, but the supplied stream fails basic strict 8b/10b invariants.
Pattern 3	Eight random-looking payload bits plus an exact 01 suffix in each 10-bit word; continuous background plus a 1 GHz comb.

Signal and analysis parameters

Quantity	Value
Input / full duration	200,000 bits per file / 20 us at 10 Gbit/s
Bit period / voltage	100 ps / -0.5 V and +0.5 V (1 V peak-to-peak)
Linear transition / impedance	20 ps / 50 ohm
Sampling	dt = 2 ps; 50 samples per bit; 10 per transition; Nyquist = 250 GHz
Segment	2,000 bits = 100,000 samples = 200 ns
Window	Kaiser beta = pi (alpha = 1); ENBW = 5.757828 MHz
FFT / grid	M = 524,288; df = 0.953674316 MHz
Resolution / averages	1/T = 5 MHz; K = 10 and 100; no segment overlap

K = 10 uses the first 20,000 bits (2 us); K = 100 uses all 200,000 bits (20 us). Both use the same segment length and window, and share the first 10 segments. The input is 200,000 bits, correcting the 2,000,000-bit statement in the earlier report.

Estimator, normalization and checks

Map each bit to bipolar level +/-A, A = 0.5 V. At a transition, start at the preceding level and reach the new level after 20 ps. The actual preceding bit is retained across segment boundaries; only the full record's start assumes preceding bit 0. No detrending or mean subtraction is applied.

$$\widehat{S}_1(f_j) = q_j \frac{\Delta t}{Z_0 \sum_{n=0}^{N-1} w_n^2} \frac{1}{K} \sum_{k=1}^{K} \left|\mathrm{FFT}_M\{w_n v_{k,n}\}_j\right|^2$$

Here N = 100,000 is the segment sample count, M is the zero-padded FFT length, and q_j = 2 at interior positive frequencies. At DC and the even FFT's Nyquist bin q_j = 1. Average squared magnitudes; averaging complex FFT coefficients would cancel random phases. Stored spectra use W/Hz; figures use 10 $\log_{10}$(1000 S) dBm/Hz.

The window power gain is U = mean(w^2) = 0.51859976; coherent gain G_c = mean(w) = 0.67107616. Broadband PSD uses U. G_c instead corrects a bin-centered tone's amplitude. ENBW = (U/G_c^2)/T = 5.757828 MHz. Both ENBW and 1/T are below 10 MHz. Zero padding gives the required sub-1-MHz display grid; it does not improve physical resolving power.

$$S_{\mathrm{NRZ},1}(f) = \frac{2A^2T_b}{Z_0}\,\mathrm{sinc}^2(fT_b)\,\mathrm{sinc}^2(fT_r), \quad f > 0$$

This is the assignment's continuous-time, one-sided independent-bit reference, with normalized sinc(x) = sin(pi x)/(pi x), for f > 0. The single DC grid point uses the undoubled value. It is a reference envelope, not the theory for coded or periodic data. The sampled waveform's exact reference differs by sinc(f dt)$^{-4}$ away from removable zeros: a limiting ratio of only 0.2867 dB at 50 GHz.

Full-band power conservation

Pattern / K	Weighted time power	PSD bin-sum power	Relative error
1 / 10	4.670993 mW	4.670993 mW	1.86e-16
1 / 100	4.670322 mW	4.670322 mW	0.00e+00
2 / 10	4.597129 mW	4.597129 mW	1.89e-16
2 / 100	4.598096 mW	4.598096 mW	1.89e-16
3 / 10	4.633671 mW	4.633671 mW	0.00e+00
3 / 100	4.636757 mW	4.636757 mW	3.74e-16

Over 0-250 GHz, sum(S_1) df equals the window-weighted time-domain power, including DC. Use the DFT bin sum, not trapezoidal endpoint half weights. The largest relative discrepancy above is 3.74e-16.

Pattern 1, K = 100: ordinary time power 4.670515 mW; sampled independent-bit expectation 4.670000 mW. Finite ramps reduce the ideal 5 mW. Over 0.1-9 GHz, RMS pointwise log error versus the sampled reference decreases from 1.417 to 0.421 dB. Zero-padded neighboring bins are correlated; these are descriptive checks.

Pattern 1: independent random NRZ

The broad envelope follows the independent random NRZ reference. Larger K reduces pointwise fluctuations while preserving the 200 ns segment's resolving power. Null spacing reflects the 100 ps UI; the transition filter first vanishes at 50 GHz.

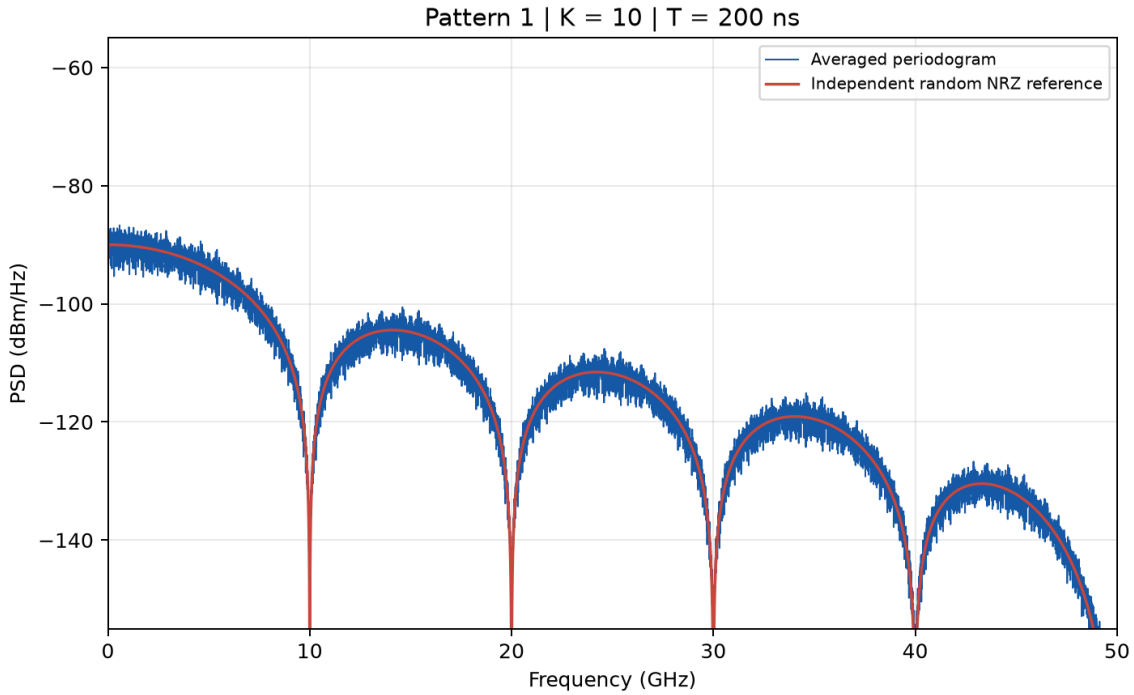


Figure 1. Pattern 1, K = 10; T = 200 ns. Blue: estimated one-sided PSD. Red: continuous-time independent random NRZ reference.

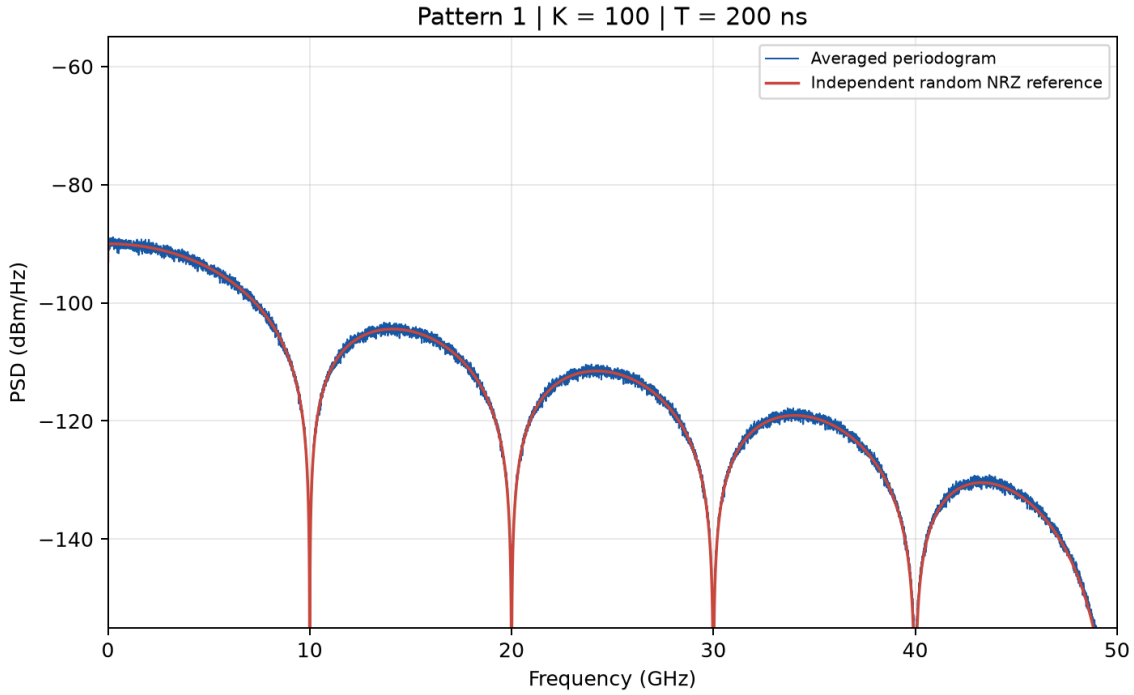


Figure 2. Pattern 1, K = 100; T = 200 ns. Blue: estimated one-sided PSD. Red: continuous-time independent random NRZ reference.

Pattern 2: intended coded example

Low-frequency power is suppressed; the average PSD over 10-100 MHz at K = 100 is -17.67 dB relative to pattern 1. Coding-related correlations explain the shaping. The bit audit on page 6 qualifies any strict 8b/10b claim.

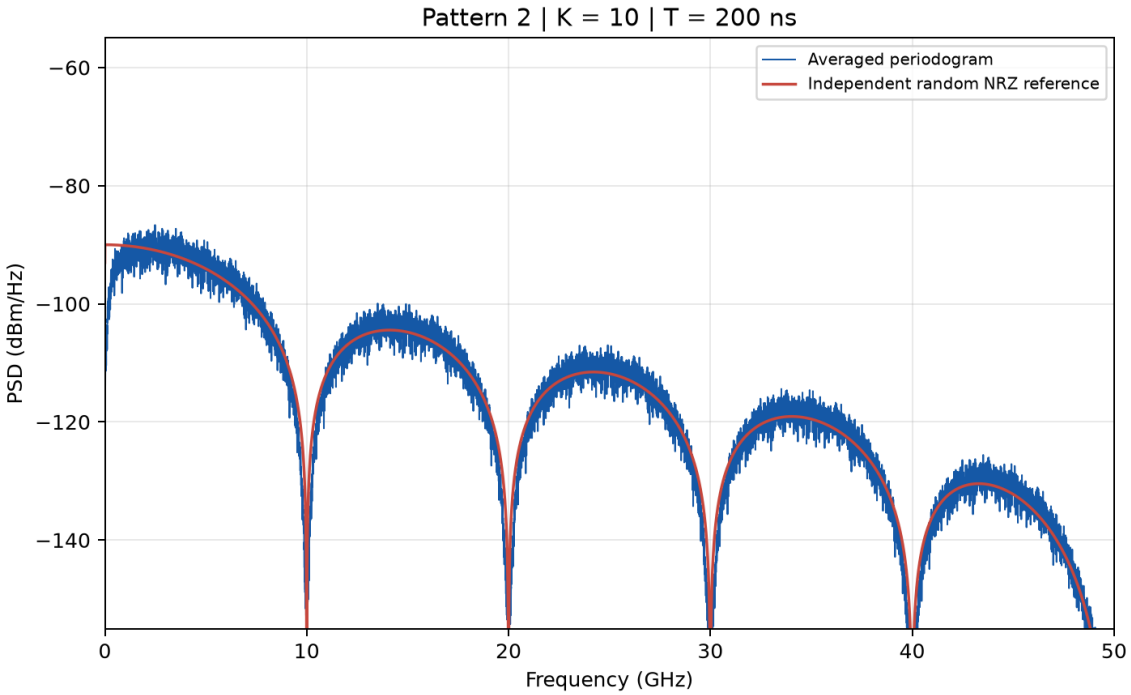


Figure 3. Pattern 2, K = 10; T = 200 ns. Blue: estimated one-sided PSD. Red: continuous-time independent random NRZ reference.

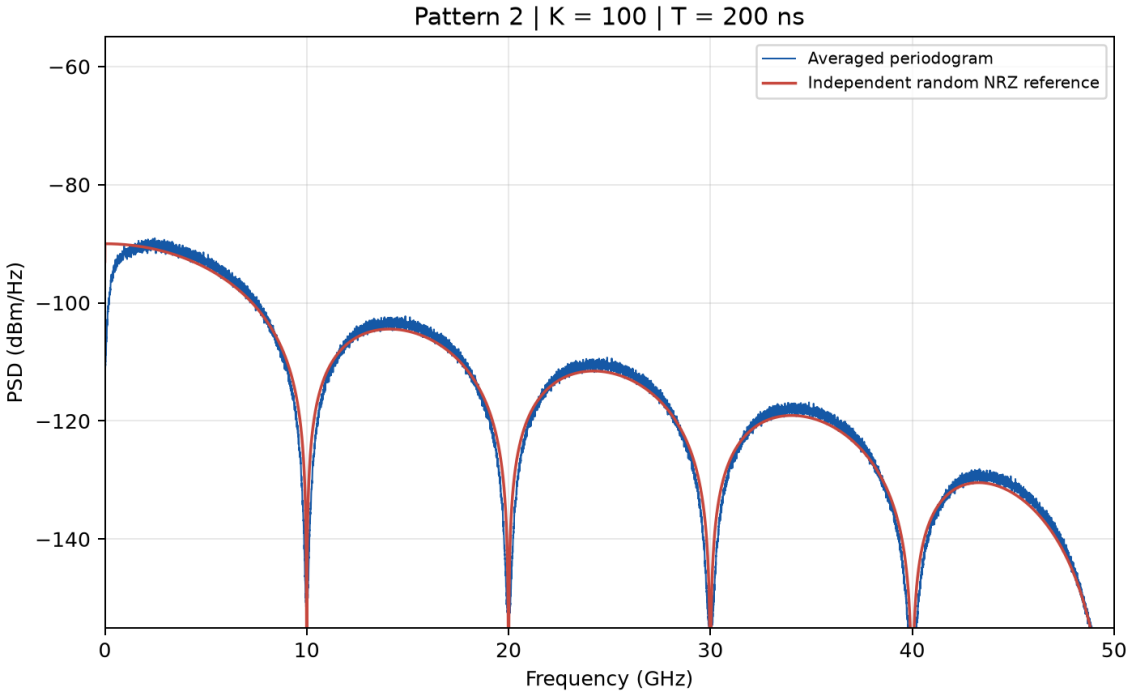


Figure 4. Pattern 2, K = 100; T = 200 ns. Blue: estimated one-sided PSD. Red: continuous-time independent random NRZ reference.

Pattern 3: hybrid random / periodic

The periodic 01 suffix adds integer-GHz lines to a random background. The 10-bit word lasts 1 ns, so its mean repeats at 1 GHz. Missing lines at 10 GHz multiples follow from the deterministic suffix and pulse zeros; the complete bitstream is not periodic.

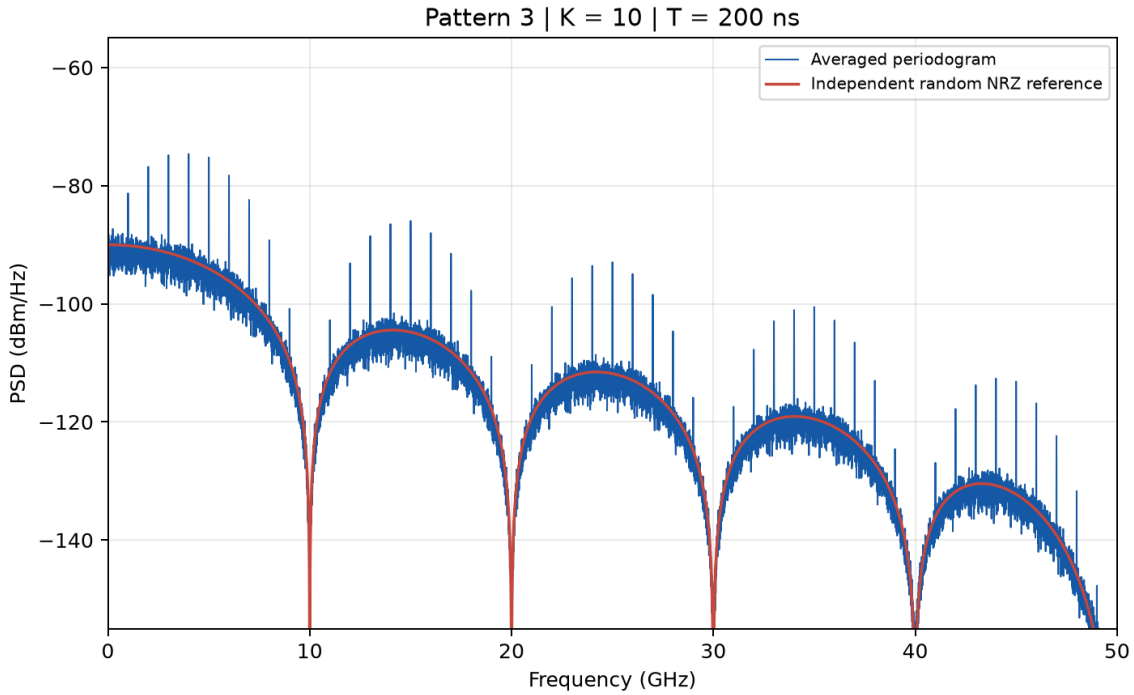


Figure 5. Pattern 3, K = 10; T = 200 ns. Blue: estimated one-sided PSD. Red: continuous-time independent random NRZ reference.

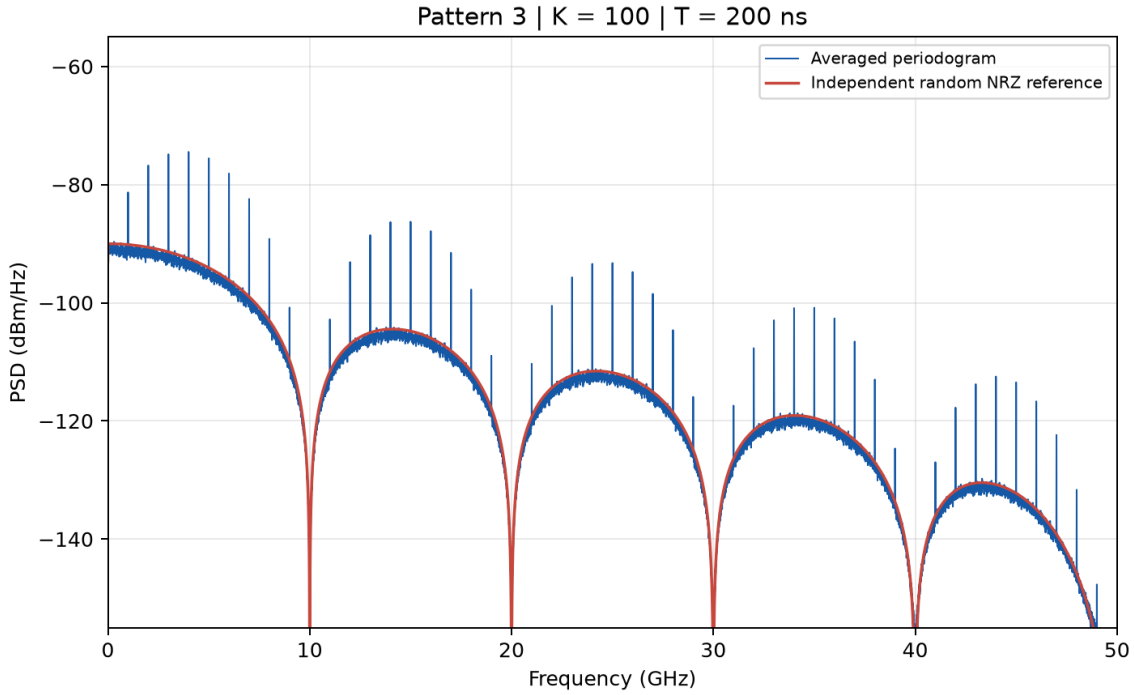


Figure 6. Pattern 3, K = 100; T = 200 ns. Blue: estimated one-sided PSD. Red: continuous-time independent random NRZ reference.

Bit-file evidence and the 8b/10b caveat

Audit (200,000 bits each)	Pattern 1	Pattern 2	Pattern 3
Ones	99,981	100,294	100,007
Maximum equal-bit run	17	6	9
Final signed bit sum	-38	588	14
Distinct aligned 10-bit words	1,024	440	256

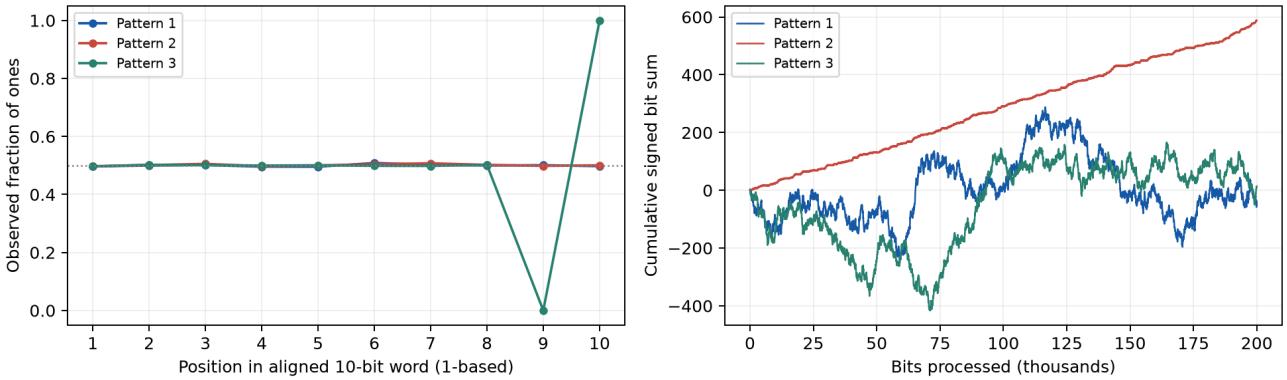


Figure 7. Left: fraction of ones by aligned word position. Right: cumulative sum of +1 for a one and -1 for a zero (display downsampled). Exact extrema and counts come from the full record.

Pattern 1 uses all 1,024 10-bit words and has no strong phase bias. Together with its PSD, this supports independent random NRZ; a finite sample cannot prove independence. Pattern 3 has 20,000 words with position 9 always 0 and position 10 always 1. All 256 possible 8-bit payload values occur. The suffix is exact; independent, equiprobable payload bits are a statistical model.

Why genuine 8b/10b suppresses low frequencies

8b/10b allows repeated bits. The relevant properties are controlled DC balance, bounded running disparity and constrained run length. These correlations reduce slow imbalance, permit AC coupling and aid clock recovery. The IBM code specifies maximum run length five and digital sum variation six [3]. AMD describes character-end running-disparity states of +1 or -1 [4]. The explanation is controlled imbalance, not an absence of consecutive zeros or ones.

What the actual supplied pattern 2 proves

Pattern 2 contains **42 runs of six bits**. Its cumulative signed bit sum spans **-2 to 590** and ends at **588**, inconsistent with a strict bounded-disparity 8b/10b stream. At the initial 10-bit alignment, 119 words have disparity +4. Alignment alone is not decisive: offset six bits removes all absolute word disparities above two, but the cumulative drift persists.

The evidence supports the assignment's intended coded example and 8b/10b-like spectral shaping. It does not validate a standards-compliant encoder or a full decoder check. The supplied bit files are preserved unchanged.

Hybrid model: background plus a comb

Under the independent, equiprobable payload model, the ten phase-conditioned bipolar means are [0, 0, 0, 0, 0, 0, 0, 0, -A, +A]. The average mean is zero, but the phase-dependent mean is nonzero and periodic. Eight positions out of ten have variance A^2, so the time-averaged continuous spectrum is 0.8 times the random NRZ reference (-0.9691 dB). The periodic mean supplies the deterministic lines.

Let $f_m = m/(10T_b) = m$ GHz. The mean waveform's Fourier-series coefficients are

$$c_m = \frac{A}{10}\,\mathrm{sinc}(f_m T_b)\,\mathrm{sinc}(f_m T_r)\,e^{-j\phi_m}\left[e^{-j2\pi m 9/10} - e^{-j2\pi m 8/10}\right]$$

The common phase phi_m contains the pulse and smoothing delays; it does not affect power. The one-sided model is

$$S_{\mathrm{hybrid},1}(f) = 0.8 S_{\mathrm{NRZ},1}(f) + \sum_{m=1}^{\infty} \frac{2|c_m|^2}{Z_0}\,\delta(f - f_m), \quad f_m = m\,\mathrm{GHz}$$

Each positive-frequency line therefore has integrated power

$$P_m = \frac{8A^2}{100 Z_0}\,\mathrm{sinc}^2(m/10)\,\mathrm{sinc}^2(m/50)\,\sin^2(\pi m/10)$$

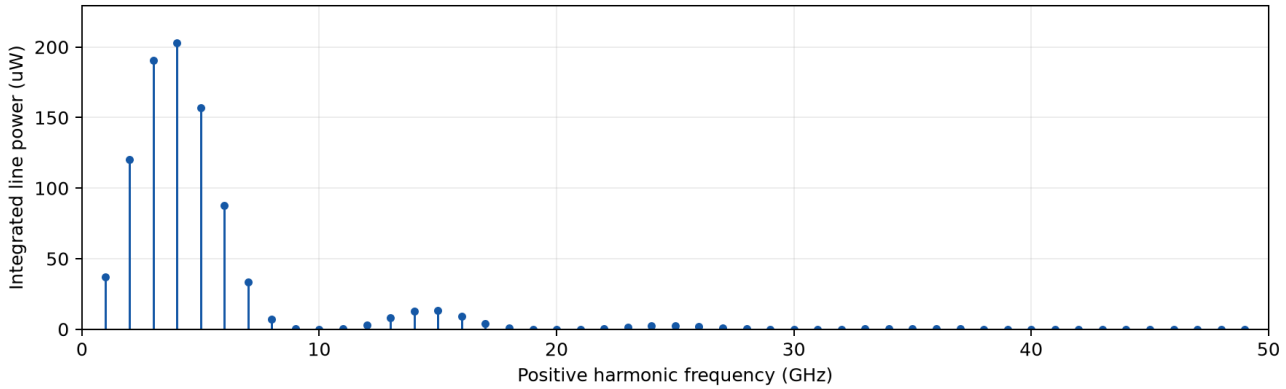


Figure 8. Analytical integrated positive-frequency line powers, not finite-window PSD peak heights. The random background is omitted from this diagnostic.

The 1 GHz fundamental follows from a 1 ns mean period. The suffix difference gives sin(pi m/10), zero at DC and m = 10, 20, ...; the NRZ pulse also vanishes at those 10 GHz multiples. The transition filter first vanishes at 50 GHz. This explains both the comb spacing and missing harmonics.

A finite window broadens each ideal delta line. A line's displayed dBm/Hz peak depends on T and the window even when its physical integrated power is fixed. Finite payload realizations fluctuate about the 0.8 continuum; observing all payload values does not prove an independent random source.

Uncertainty, reproduction and references

Uncertainty and interpretation

For independent periodograms of a Gaussian-like continuous random spectrum, the relative standard deviation is approximately 1/sqrt(K): 31.6% at K = 10 and 10.0% at K = 100. The small-error dB approximations are 1.37 and 0.43 dB. This heuristic is not a line-amplitude confidence interval and does not apply unchanged to nulls, DC, deterministic lines or correlated coded data.

Window leakage limits apparent null depth and broadens narrow lines. Neighboring zero-padded bins are correlated. The continuous reference assumes an ideal pulse model; the sampled reference quantifies the small discretization difference. The bit audit identifies the supplied pattern 2's strict-code inconsistencies; no source data are corrected to hide them.

Reproduce this report

Step	Command from the homework directory
Install dependencies	python -m pip install -r requirements.txt -r report/requirements.txt
Regenerate analysis	python src/hw4.py
Run verification	python -m unittest discover -s tests -v
Independent SciPy audit	python tests/verify_reference.py
Rebuild PDF and Markdown	python report/build_report.py
Delivery integrity check	python tests/verify_delivery.py

All 20 tests pass. A separate waveform construction and SciPy density periodogram reproduce all six saved spectra (262,145 bins each) without calling the rewritten estimator. Worst relative L2 error: 2.533e-15; worst full-band power discrepancy: 3.773e-16. Detailed results: results/verification.json.

summary.json stores configuration, window gains, bit statistics and six power checks. spectra.npz stores the full frequency grid, all linear PSD estimates and both random-reference curves. pattern_audit.json stores a separate audit including SHA-256 checksums and all ten word alignments. All inputs and generated paths are package-relative. The PDF title is generic and the author metadata is empty.

References

[1] Course assignment, *Homework: Spectrum of digital signals*, pp. 1-2. Included as instructions/HW4_spectrum_of_digital_signals.pdf.

[2] Course lecture, *Spectrum of digital signals*, pp. 2-8, 15-16 and 40-43. PSD definitions, averaging, periodic/random spectra and window gains. Used as a read-only teaching reference.

[3] A. X. Widmer and P. A. Franaszek, *Transmission code for high-speed fibre-optic data networks*, 17 March 1983. IBM Research: original code reference.

[4] AMD, *Versal Adaptive SoC GTY and GTYP Transceivers Architecture Manual (AM002): Running Disparity*. Official running-disparity documentation.

[5] *8b/10b encoding*, Wikipedia. Assignment-requested overview. Run-length and disparity claims are additionally checked against [3]-[4].